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1. Critical Viscosity Exponent
A self-consistent mode coupling calculation of the critical viscosity exponent for classical fluids is performed and extended by the inclusion of some higher loop contributions. There are 18 different Feynman graphs corresponding to the graphs hacking 3 loops. These need to be combined into a smaller number of multiple integrals which then have to be evaluated.

2. Specific Heat in Confining Geometry

a) This is the subject of CheX. It poses a serious challenge to the theory, particularly in the passage through the lambda point, where the onset of 2-dimensionality is expected. b) Fixing the Scale. The theory of confinement is compared with the earth-based data of the CHEX findings in our publication in the Journal of Low Temperature Physics (see

Task Description:	bibliography). Although this verification of the theory is entirely satisfactory, it is incomplete in the employment of a free parameter, the geometrical scale factor for the order parameter fluctuations. Our current theoretical work is being devoted to using the idea of light scattering as a means of independently fixing the scale. To carry this out, we are finding that it may be necessary to extend the light scattering to the experimentally very difficult infared region. 3. Thermal Conductivity There is a strong incentive to move out of the quasiscaling region and closer to the lambda point and the true asymptotic region by a flight experiment (DYNAMX, R. Duncan, P. I.). In BEST (G. Ahlers, P. I.) attention is given to the effect of confinement on the thermal conductivity. 4. Superconductivity and Cosmological Analogies Because the universe is thought to have undergone some phase transitions in its earlier stages of evolutions, it is useful to compare these phase transitions with those that are familiar in condensed matter physics. The task objectives are to develop detailed theories for the flight experiments CVX (critical viscosity experiment), and CheX, and to provide theoretical background and support for the manifested low temperature flight experiments BEST and DYNAMX, as well as for other proposed experiments. The task objectives can be grouped into four categories as follows: 1. Critical viscosity exponent 2. Specific heat in confining geometry 3. Thermal conductivity of superfluid helium in confining geometry 4. Superconductivity and cosmological analogies 1. Critical Viscosity Exponent A self-consistent mode coupling calculation of the critical viscosity exponent for classical fluids is performed, including the memory effect and the vertex corrections. The incorporation of the memory effect is through a self-consistency procedure that evaluates the order parameter and shear momentum relaxation rates at non-zero frequencies, thereby taking their frequency dependence into account. This approach offers considerable simplification and efficiency in the calculation. The vertex corrections are also demonstrated to have significant effects on the numerical values for the critical viscosity exponent, in contrast to some previous theoretical work, which indicated that the vertex corrections tend to cancel out from the final result. By carrying out all of the integrations analytically, we have succeeded in tracing the origin of this discrepancy to an error in the earlier work. 2. Specific Heat in Confining Geometry It is particularly important to extend the theory below the lambda point. 3. Thermal Conductivity The good agreement between theory and the ground-based measurements by Kahn and Ahlers of the effect of confinement on the thermal conductivity indicates that it will be a reliable predictor for the ISS experiment BEST. 4. Superconductivity and Cosmological Analogies We have explored the idea that the enormous amount of energy released in a gamma ray burst might come from the crossing of two cosmic strings. The laboratory analogy of two crossed vortices in superfluid Helium-four is being studied with the goal of developing some intuitive feeling for this kind of process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	I. Gamma Ray Bursts Since the summer of 2004, when NASA launched the gamma-ray observatory, “Swift”, gamma-ray bursts have been observed and registered at an average of approximately two per day. Because of their very large red shift, it follows that these sources are extremely far away, according to Hubble’s law, for the intensity to be sufficient strong to nevertheless be detectable and measurable in the vicinity of the earth, the intrinsic strength can be inferred to be of such a greater magnitude as heretofore to have ever been observed. We are considering a possible explanation for this by invoking cosmic strings, thought to have been created in the early universe, and to contain enormous amounts of energy in their cores. We are studying the release of some of this energy when two cosmic strings cross one another, and are developing a prediction for the time signature of this release of gamma-ray energy. II. Cosmic String Analogy in Superfluid Helium The extremely high-energy yield that is observed in the gamma ray bursters has impelled theoretical astrophysicists to try to find an explanation for the source in some unusual natural phenomenon. One candidate is based upon the idea that one or more phase transitions occurred in the early universe, leading to the formation of regions of differing order parameter. In the boundaries of these regions it is presumed that topological singularities could have been trapped—the so-called “cosmic strings”, which are expected to have stored enormous energy per unit length. These entities would necessarily be stable, except when two of them happen by accident to cross, causing them to release some of their energy. With the goal of developing some intuitive feeling for such a process, we have studied the analogous situation of the crossing of two vortices in superfluid Helium-four. It remains to estimate the sensitivity required for detecting the sonic pulses that are expected to be generated by the crossings. III. Dynamic Scaling Theory in a Classical Fluid In a classical fluid near its phase transition there are large amplitude fluctuations in two strongly interacting fluids. These

Task Progress:	are the transverse hydrodynamic modes and the scalar order parameter. The latter is the mass density in a one-component fluid or the concentration in a binary liquid. We have recently completed in great detail a calculation of the numerical value of the critical viscosity exponent, which describes quantitatively the manner in which the viscosity diverges as the temperature is lowered toward the critical temperature. The calculation was carried out by a step-by-step application of Feynmann diagrams in which each interaction of the modes is represented by a point with which lines corresponding to the modes are attached. Because of the variety in which the attachments can be made, the "tree" diagrams proliferate, which necessitates the inclusion of a quite large number of Feynmann diagrams. There are 18 such diagrams of the 3-loop type, each corresponding to a multidimensional integration. In the effort to establish a simpler, more straightforward method of carrying out the calculation we have returned to a method that we employed many years in the study of the critical dynamics at and near the lambda transition. This used an identity based on the requirement of thermal equilibrium for combining and reducing the number of tree graphs. We believe that the introduction of a similar identity will help in making the evaluation of the critical viscosity exponent more cogent and compelling. IV. Viscosity in a Two-Dimensional Fluid The pioneering space shuttle measurement of the critical viscosity by Berg, Moldover, and Zimmerli (CVX) was based on the observation of the viscous drag on a vibrating mesh of very thin cylinders. Such a cylinder can be treated as a two-dimensional analogue of Stokes' Law for the drag on a moving sphere. There is, however, no true 2-D analogue for uniform motion of an infinitely long cylinder. The drag depends logarithmically on l, the length of the cylinder. We have found that the cutoff on the logarithmic divergence is no longer l for an oscillating cylinder, but is instead determined by the viscous penetration depth. This, in turn, is a function of the frequency of the oscillation, which thereby leads to a finite value for the drag.
Bibliography Type:	Description: (Last Updated: 04/30/2008)
Articles in Peer-reviewed Journals	Hao H, Ferrell RA, Bhattacharjee JK. "Critical viscosity exponent for classical fluids." Phys Rev E Stat Nonlin Soft Matter Phys. 2005 Feb;71(2 Pt 1):021201. PMID: 15783317 , Feb-2005